

Kondapally Madhavi

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Hyderabad, Telangana State - 500049, India

RESEARCH INTERESTS

- Autonomous Driving Perception and Advanced Driver Assistance Systems (ADAS)
- Robust Computer Vision under Adverse Weather Conditions
- Vision-Language Models, Vision-Language-Action Models, and Multimodal AI
- Large Language Models and AI Agents
- Medical AI and Medical Image Analysis

ACADEMIC BACKGROUND

- **Indian Institute of Technology Hyderabad (IIT Hyderabad)** *Aug 2021 - Mar 2026*
Doctor of Philosophy (PhD)
Hyderabad, India
 - **Department:** Computer Science & Engineering
 - **Thesis:** Scene Perception for Autonomous Vehicle Technology in Transitional Weather Conditions
 - **Supervisor:** Prof. C Krishna Mohan
 - **CGPA:** 9.25/10.00
- **Jawaharlal Nehru Technological University, Hyderabad, India (JNTU Hyderabad)** *Sep 2008 - Dec 2010*
Master of Technology (MTech)
Hyderabad, India
 - **Department:** Computer Science & Engineering
 - **Percentage:** 73 %
- **Kakathiya University, Warangal, India** *June 2002 - Apr 2006*
Bachelor of Technology (BTech)
Warangal, India
 - **Department:** Information Technology
 - **Percentage:** 73 %

INDUSTRY EXPERIENCE

- **Solugenix India Private Limited** *Feb 2026 - Present*
Senior Data Scientist
Hyderabad, India
 - Fine-tuned Small Language Models (SLMs) and Large Language Models (LLMs) for enterprise AI applications.
 - Developed LLM- and RAG-based Intelligent Document Question Answering (QA) systems for enterprise knowledge retrieval.
 - Built AI agents to automate enterprise workflows, including account unlocking, employee onboarding, and knowledge assistance.
 - Designed automated evaluation and testing frameworks to assess model accuracy, reliability, robustness, and performance.
 - Collaborated with cross-functional teams to develop, deploy, and maintain scalable AI solutions.
 - Documented system architectures, experiments, and model evaluation results to ensure reproducibility and knowledge sharing.

TEACHING EXPERIENCE

- **B.V. Raju Institute of Technology, Narsapur (BVRIT Narsapur)** *Feb 2015 - Aug 2021*
Assistant Professor
Hyderabad, India
 - **Department:** Information Technology
- **Marri Laxman Reddy Institute of Technology (MLRITM Hyderabad)** *Dec 2011- Oct 2012*
Assistant Professor
Hyderabad, India
 - **Department:** Computer Science and Engineering
- **TRR Engineering College (TRREC, Hyderabad)** *Dec 2010 - Dec 2011*
Assistant Professor
Hyderabad, India
 - **Department:** Computer Science and Engineering

- [J.1: IEEE T-ITS] Kondapally Madhavi, K. Naveen Kumar, C. Krishna Mohan, **Towards a Transitional Weather Scene Recognition Approach for Autonomous Vehicles**. *IEEE Transactions on Intelligent Transportation Systems*, Vol. 25, Issue 6, pp. 5201-5210, 2024. [IF: 8.5]
- [J.2: Elsevier PR] Kondapally Madhavi, K. Naveen Kumar, C. Gayathri, **TWFNet: Introducing Transitional Weather Conditions for Autonomous Driving with a Spatio-temporal Forecasting Network**. *Pattern Recognition (Elsevier)*, pp. 112-154, 2025. [IF: 7.6]
- [J.3: TMLR] Kondapally Madhavi, K. Naveen Kumar, C. Krishna Mohan, **Eyes on the Road, Words in the Changing Skies: Vision-Language Assistance for Autonomous Driving in Transitional Weather**. *TMLR* 2026.
- [C.1: WACV] Kondapally Madhavi, K. Naveen Kumar, C. Krishna Mohan, **CaRS: A Causal Intervention Segmentation Framework and Benchmark Dataset for Autonomous Driving under Transitional Weather Conditions**. *Winter Conference on Applications of Computer Vision (WACV) 2026*, Arizona, USA.
- [C.2: IJCNN] Kondapally Madhavi, K. Naveen Kumar, C. Krishna Mohan, **Object Detection in Transitional Weather Conditions for Autonomous Vehicles**. In *International Joint Conference on Neural Networks*, pp. 1-8. IEEE. June, Yokohama, Japan 2024
- [C.3 Springer] RaviKiran Ramaraju, Kondapally Madhavi, G. Ravi, **Sentimental Analysis on Twitter Data Using Hadoop with Spring Web MVC**. *Intelligent System Design. Advances in Intelligent Systems and Computing* (Springer Nature), vol 1171. Springer, Singapore, pp. 265–273, 2020
- [W.1: ICPR] Kondapally Madhavi, K. Naveen Kumar, C. Krishna Mohan, **TransWardX: An Explainable Black-box Object Detection Attack for Autonomous Driving in Transitional Weather Conditions**, *International Conference on Pattern Recognition*. Springer Nature Switzerland, 2024. *Lecture Notes in Computer Science*, vol 15619.
- [W.2: IEICE] Kondapally Madhavi, C. Krishna Mohan, **Weather Scene Perception for Autonomous Vehicles**. In *International Workshop on Computer Vision and Artificial Intelligence*, IEICE proceedings, Japan, pp.61-64
- [S.1: TMLR] Kondapally Madhavi, K. Naveen Kumar, C. Krishna Mohan, **TransWeatherNet: End-to-End Restoration Framework and Benchmark for Robust Autonomous Vehicle Perception under Transitional Weather Conditions**. *TMLR* 2026
- [S.2: TMLR] Ishu Priya, K. Naveen Kumar, Kondapally Madhavi, C. Krishna Mohan, **AO-Fed: A Large-Scale Blood Cell Dataset with Semantic Attribute Orchestration for Class-Disjoint Label Skew Federated Learning**. *TMLR* 2026
- [S.3: WACV] Aashish Singh, Kondapally Madhavi, C. Krishna Mohan, **Retrieval-Augmented Vision-Language Planning for Autonomous Driving: Integrating Episodic Memory with Scene-Grounded Trajectory Prediction**. *WACV* 2026
- [S.4: AAAI] Kondapally Madhavi, Indra Sonowal, K. Naveen Kumar, Mohsen Guizani, **AgentDriveTW: An Agentic Vision-Language-Action Framework for Autonomous Driving Under Transitional Weather**. *AAAI* 2027

PATENTS

- Kondapally Madhavi, K. Naveen Kumar, C. Krishna Mohan, Sobhan Babu, "System And Method For Performing Adaptive Object Detection In An Autonomous Vehicle System", Indian Patent No. 572776. Indian Patent Office. Filed Jan 7, 2025. Granted Oct 28, 2025. Application No. 202541001505.
- Kondapally Madhavi, K. Naveen Kumar, C. Krishna Mohan, Sobhan Babu, "System and method for generating weather transition data for autonomous vehicle training", Indian Patent No. 582123. Indian Patent Office, Filed Jan 03, 2025. Granted Feb 27, 2026. Application no. 202541000718.
- Kondapally Madhavi, K. Naveen Kumar, C. Krishna Mohan, "System and method for visual segmentation for autonomous vehicles under transitional weather conditions", Indian Patent Office, Application no. 202541129186, Dec, 19, 2025

FUNDED PROJECTS

- **Medicine from the sky** Sep 2021 - Dec 2021
Project title: Design and Development of AI-based real-time light-weight system medical drone delivery
 - **Funded by:** Bold and Unique Ideas Leading to Development (BUILD), IITH
 - Designed and implemented a lightweight AI-based scene perception module for autonomous medical drone delivery, integrating QR code detection, obstacle detection, and monocular depth estimation for fast and reliable navigation.
- **Advanced drive assistance system** Mar 2025 - Dec 2025
Project title: Development of Road Element Segmentation and Language-Guided Vision Models for Autonomous Driving
 - **Funded by:** TiHAN IIT Hyderabad
 - Researched and developed robust semantic segmentation algorithm for reliable road scene perception in challenging weather environments.

- Designed multimodal vision-language dataset and trained Vision-Language Model (VLM) for scene-level reasoning and question answering.
- Developed an integrated perception framework combining semantic segmentation and vision-language understanding for ADAS applications.
- Optimized model architectures to improve computational efficiency, inference speed, and deployment feasibility for real-world autonomous vehicles.
- Conducted extensive experimentation, performance evaluation, and documentation to validate model effectiveness and support project deliverables.

TECHNICAL SKILLS

- Machine learning, deep learning, supervised and unsupervised learning, computer vision, Document Retrieval, and SLM/LLM
- **Programming & Libraries:** Python, TensorFlow, PyTorch, and OpenCV

ADDITIONAL INFORMATION

- **Teaching Assistant** for the below courses offered by Prof. C Krishna Mohan (PhD supervisor) at IIT Hyderabad
 - * CS6450 - Visual Computing
 - * CS6140 - Video Content Analysis
 - * CS6170 - Computer Vision for Autonomous Vehicle Technology
 - * CS6870 - Surveillance Video Analytics
- **External Reviewer**
 - * IEEE International Joint Conference on Neural Networks (2024)
 - * IEEE Intelligent Transportation Systems Conference (2024)
 - * Elsevier Neurocomputing (2023)
- Student member of International Neural Network Society (INNS)

REFERENCES

1. **Dr. C Krishna Mohan (PhD Supervisor)**

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Indian Institute of Technology Hyderabad
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1. **Dr. Sobhan Babu**

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1. **Dr. Sathya Peri**

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